

VIRTUSU, Estonia

WMO: 261280

Lat: 58.5728N

Lon: 23.5136E

Elev: 12

StdP: 14.69

Time Zone: 2.00 (EUE)

Period: 99-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-0.4	5.3	-5.0	4.2	0.7	0.1	5.5	6.0	25.4	29.1	21.8	29.4	4.4	100	0.579

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.1	79.1	68.2	75.8	66.5	73.0	65.0	69.9	76.7	67.9	74.2	66.1	71.8	6.0	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.3	100.5	74.1	65.4	93.8	72.0	63.5	87.9	70.0	33.9	76.5	32.3	74.2	30.8	72.0	75.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
20.9	18.0	15.7	-4.1	83.0	9.1	4.1	-10.6	85.9	-15.9	88.4	-21.0	90.7	-27.6	93.7		
			-4.7	71.8	8.0	2.4	-10.4	73.5	-15.1	74.8	-19.6	76.2	-25.3	77.9		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	44.7	26.4	25.5	30.1	40.4	52.0	59.1	65.2	63.7	55.6	45.7	38.4	32.6
	DBStd	15.44	10.19	9.64	7.38	6.15	6.61	5.06	4.84	4.41	5.01	6.67	6.64	8.51
	HDD50	3443	732	687	616	294	50	1	0	0	14	162	347	540
	HDD65	7535	1197	1107	1081	737	408	189	55	77	283	599	797	1005
	CDD50	1495	0	0	0	7	110	273	473	423	181	28	0	0
	CDD65	113	0	0	0	0	4	11	63	35	0	0	0	0
	CDH74	498	0	0	0	0	22	52	290	133	1	0	0	0
	CDH80	70	0	0	0	0	2	5	44	19	0	0	0	0
Wind	WSAvg	7.8	8.3	7.8	7.8	7.8	7.3	7.6	6.9	6.7	7.1	7.9	8.7	9.2
Precipitation	PrecAvg	24.7	2.0	1.6	1.3	1.2	1.3	2.2	2.5	3.0	2.2	2.7	2.5	2.1
	PrecMax	35.9	4.4	3.6	2.5	2.6	2.7	3.7	5.6	6.6	4.8	5.3	4.4	5.5
	PrecMin	13.1	0.4	0.2	0.1	0.2	0.2	0.6	0.1	0.2	0.2	0.4	0.0	0.5
	PrecStd	4.7	0.9	0.8	0.7	0.7	0.7	0.9	1.2	1.5	1.1	1.2	1.1	1.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	42.9	40.3	50.1	62.6	76.7	79.8	84.8	82.1	69.9	60.3	50.2	47.4
		MCWB	41.2	37.5	41.4	51.5	61.6	64.8	69.4	69.5	62.6	56.2	48.4	45.3
	2%	DB	40.4	37.8	43.9	57.5	71.3	74.3	80.1	77.4	67.0	57.6	48.6	45.1
		MCWB	39.6	36.3	38.9	47.6	59.6	64.0	69.2	67.3	60.8	54.9	47.0	43.8
	5%	DB	38.5	36.3	40.2	53.7	66.9	70.8	77.0	74.1	64.9	56.1	47.5	43.6
		MCWB	37.3	35.3	37.1	46.3	58.0	62.1	68.2	65.9	59.6	53.9	46.0	42.5
	10%	DB	36.7	35.1	37.8	50.0	62.7	67.9	74.2	71.6	63.0	54.3	46.4	41.6
		MCWB	35.7	34.2	35.5	43.9	55.2	60.5	66.6	64.5	58.4	52.4	44.9	40.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	42.0	38.2	43.7	53.0	64.8	68.2	72.5	71.8	63.9	57.3	48.9	45.9
		MCDB	42.9	39.5	47.9	59.4	71.1	75.3	79.8	79.1	68.0	58.4	49.6	46.9
	2%	WB	39.7	36.6	39.4	49.5	61.6	65.6	70.7	69.2	61.9	55.6	47.5	44.1
		MCDB	40.3	37.5	42.7	55.7	68.7	72.5	77.8	75.2	65.7	57.3	48.4	44.8
	5%	WB	37.4	35.5	37.6	46.8	58.9	63.3	69.0	67.1	60.3	54.1	46.3	42.5
		MCDB	38.2	36.2	39.8	52.2	65.8	69.4	75.7	72.7	63.8	55.7	47.3	43.4
	10%	WB	35.8	34.5	36.0	44.5	55.8	61.1	67.2	65.3	59.0	52.7	45.0	40.5
		MCDB	36.6	35.1	37.4	48.8	61.7	66.8	73.4	70.3	62.2	54.2	46.1	41.4
Mean Daily Temperature Range	5% DB	MDBR	7.2	7.9	9.9	10.9	12.1	11.0	11.1	11.5	9.8	8.0	5.6	5.9
		MCDBR	5.0	6.0	11.1	17.2	18.6	16.3	15.8	16.1	12.2	8.3	5.5	5.0
		MCWBR	4.8	5.1	7.8	10.5	10.3	8.8	8.3	8.8	7.8	6.4	5.1	4.8
	5% WB	MCDBR	4.9	5.6	10.0	14.7	16.5	14.1	13.9	13.6	10.4	7.2	5.4	4.6
MCWBR		4.8	4.8	7.3	9.7	9.7	8.4	8.3	8.3	7.4	6.1	5.3	4.6	
Clear-Sky Solar Irradiance	taub	0.287	0.314	0.337	0.374	0.368	0.367	0.396	0.390	0.358	0.340	0.324	0.283	
	taud	2.169	2.196	2.248	2.307	2.402	2.440	2.380	2.394	2.465	2.404	2.232	2.139	
	Ebn at Noon	166	221	253	263	274	276	264	256	247	214	150	126	
	Edh at Noon	16	24	31	35	34	33	34	31	25	20	15	11	
All-Sky Solar Radiation	RadAvg	93	272	760	1279	1801	1901	1774	1397	874	394	114	62	
	RadStd	16	51	75	143	162	103	138	125	102	50	22	8	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (41 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page